# 9 Weeks 9–10: Unmeasured Confounding and Sensitivity Analysis

## 9.1 Motivation

Up to Week 8, our identification strategy relied on the strong ignorability assumption

$$(Y(1), Y(0)) \perp Z \mid X,$$

which states that, conditional on observed covariates $X$, treatment assignment $Z$ is as good as random. This assumption is powerful, but in observational studies it is rarely guaranteed. There are often unobserved factors—motivation, risk preference, health-seeking behavior, family support, genetic predisposition—that influence both treatment and outcome. Once we admit this possibility, we must confront the central difficulty:

> *What can we still learn when treatment assignment may depend on unmeasured covariates?*

In these notes, we develop a formal understanding of unmeasured confounding, examine its consequences, and introduce two broad strategies for strengthening causal inference:

1. using multiple control groups or negative controls to probe the presence of unmeasured confounding, and
2. conducting sensitivity analyses to quantify how strong unmeasured confounding would need to be to overturn our conclusions.

Our goal is not to eliminate unmeasured confounding—which is impossible without additional assumptions—but to reason transparently about its potential impact.

### 2.1 A Linear Example

Suppose the true outcome model is

$$Y_i = \tau Z_i + \beta X_i + \gamma U_i + \varepsilon_i, \qquad E[\varepsilon_i \mid Z_i, X_i, U_i] = 0,$$

but we estimate the misspecified regression

$$Y_i = \alpha + \tilde{\tau} Z_i + \tilde{\beta} X_i + \tilde{\varepsilon}_i,$$

omitting $U_i$. The omitted variable bias formula yields

$$\boxed{\tilde{\tau} \approx \tau + \gamma \cdot \frac{\mathrm{Cov}(Z_i, U_i \mid X_i)}{\mathrm{Var}(Z_i \mid X_i)}.}$$

The bias is the product of two forces:

- $\gamma$: the effect of $U$ on $Y$,
- $\mathrm{Cov}(Z, U \mid X)$: the association between $U$ and $Z$ after adjusting for $X$.

If either term is zero, the bias vanishes; otherwise, the regression coefficient is systematically shifted.

### 9.1.1 A Concrete Illustration

Consider a job training program. Let $U$ represent "motivation," which is unobserved. More motivated individuals are more likely to enroll ($U \to Z$) and also tend to have higher earnings regardless of treatment ($U \to Y$). Even after adjusting for $X$ (age, education, prior earnings), the treated group may still be more motivated on average. The estimated treatment effect will then mix the true causal effect with the effect of motivation.

This example highlights the fundamental challenge: no adjustment on $X$ alone can remove bias from an unobserved confounder $U$.

## 9.2 Strengthening Causal Inference Using Multiple Control Groups

If $U$ is unobserved, we cannot adjust for it directly. However, we can design comparisons that are differentially sensitive to $U$ and examine whether the estimated effect changes. This idea goes back to Campbell and Stanley (1963) and Rosenbaum (1987).

### 9.2.1 Setup

Suppose we have:

$$T \quad \text{(treated group)}, \qquad C_1 \quad \text{(control group 1)}, \qquad C_2 \quad \text{(control group 2)}.$$

We suspect that $U$ differs between $C_1$ and $C_2$ even after adjusting for $X$. For example:

- $C_1$: individuals who applied but were not selected,
- $C_2$: individuals who never applied.

Applicants may be more motivated than non-applicants, so $E[U \mid C_1] > E[U \mid C_2]$.

### 3.2 Diagnostic Comparison

Define

$$\hat{\tau}_{T,C_1}, \qquad \hat{\tau}_{T,C_2},$$

after balancing on $X$ within each comparison. If unmeasured confounding is present, the two estimates may differ:

$$\hat{\tau}_{T,C_1} \neq \hat{\tau}_{T,C_2}.$$

Agreement between the two estimates increases credibility; disagreement signals sensitivity to unmeasured confounding.

### 3.3 Negative Controls

A negative control outcome $Y^{NC}$ is an outcome that the treatment cannot plausibly affect. If, after adjusting for $X$, we still observe an association between $Z$ and $Y^{NC}$, this suggests unmeasured confounding. Negative controls do not solve the problem but provide a valuable diagnostic.

## 9.3 Negative Controls: A Diagnostic Tool for Unmeasured Confounding

In the previous sections, we developed formal sensitivity analysis tools that quantify how large an unmeasured confounder would need to be to overturn our conclusions. These methods are powerful, but they require us to posit a parametric bound on hidden bias. There is, however, another complementary strategy that does not rely on specifying a sensitivity parameter. This strategy uses what are known as *negative controls*. Negative controls do not eliminate unmeasured confounding, but they provide a diagnostic window into whether such confounding is likely to be present and in what direction it might operate.

The idea of negative controls is conceptually simple. A negative control is a variable—either an exposure or an outcome—that should not be causally affected by the treatment under study. If we nevertheless observe an association between the treatment and this negative control, the association cannot be causal. It must arise from bias, typically unmeasured confounding, measurement error, or selection effects. Thus, negative controls serve as a kind of "placebo test" for causal inference.

### 9.3.1 Negative Control Outcomes

A *negative control outcome* is an outcome $Y^{\mathrm{NC}}$ that the treatment $A$ cannot possibly affect. Formally, for all units,

$$Y^{\mathrm{NC}}(1) = Y^{\mathrm{NC}}(0).$$

This is a strong assumption, but in many applications it is justified by substantive knowledge. For example, if we are studying the effect of a job training program on earnings, a negative control outcome might be the participant's height. The program cannot change height, so any association between treatment and height must be due to confounding.

If we observe

$$\mathbb{E}[Y^{\mathrm{NC}} \mid A = 1] \neq \mathbb{E}[Y^{\mathrm{NC}} \mid A = 0],$$

this signals that the treated and control groups differ in ways unrelated to the treatment. The same unmeasured factors that create this spurious association may also bias the estimated effect of $A$ on the primary outcome.

Negative control outcomes therefore provide a qualitative diagnostic: they do not quantify the magnitude of hidden bias, but they reveal its presence.

### 9.3.2 Negative Control Exposures

A *negative control exposure* is a variable $A^{\mathrm{NC}}$ that cannot possibly affect the outcome $Y$. Formally,

$$Y(a, a^{\mathrm{NC}}) = Y(a, a'^{\mathrm{NC}}) \quad \text{for all } a^{\mathrm{NC}}, a'^{\mathrm{NC}}.$$

If $A^{\mathrm{NC}}$ is associated with $Y$, this association must arise from confounding. For example, in environmental epidemiology, researchers sometimes use future exposure as a negative control exposure. Tomorrow's pollution level cannot affect today's hospitalization risk, so any association between future pollution and current health outcomes indicates confounding or measurement error.

Negative control exposures are particularly useful when the primary exposure is difficult to measure or is subject to time-varying confounding.

### 9.3.3 A Simple Mathematical Illustration

Suppose the true data-generating process is

$$Y = \tau A + \gamma U + \varepsilon,$$

where $U$ is an unmeasured confounder. Let $Y^{\mathrm{NC}}$ be a negative control outcome satisfying

$$Y^{\mathrm{NC}} = \delta U + \eta.$$

Because $A$ does not affect $Y^{\mathrm{NC}}$, any observed association between $A$ and $Y^{\mathrm{NC}}$ must arise from the shared confounder $U$. Indeed,

$$\mathbb{E}[Y^{\mathrm{NC}} \mid A = 1] - \mathbb{E}[Y^{\mathrm{NC}} \mid A = 0] = \delta\big(\mathbb{E}[U \mid A = 1] - \mathbb{E}[U \mid A = 0]\big).$$

If this difference is nonzero, then

$$\mathbb{E}[U \mid A = 1] \neq \mathbb{E}[U \mid A = 0],$$

which implies that the naive estimate of $\tau$ is biased. Thus, negative controls provide a direct empirical probe into the structure of unmeasured confounding.

### 9.3.4 Empirical Example: Vaccines and Birth Outcomes

A classic example comes from perinatal epidemiology. Researchers have studied whether influenza vaccination during pregnancy affects birth outcomes such as birth weight or preterm birth. A concern is that women who choose to be vaccinated may differ from those who do not in terms of health-seeking behavior, socioeconomic status, or access to care.

A negative control outcome in this setting is the infant's congenital anomalies. Influenza vaccination cannot cause congenital anomalies, because these anomalies arise early in pregnancy, before vaccination typically occurs. If vaccinated mothers have infants with fewer congenital anomalies, this suggests that the vaccinated group is healthier or more advantaged in ways unrelated to the vaccine. This imbalance signals that the estimated effect of vaccination on birth outcomes may be confounded.

### 9.3.5 Empirical Example: Air Pollution and Mortality

In environmental epidemiology, researchers often study the effect of short-term air pollution exposure on mortality. A negative control exposure commonly used is *future* pollution. Tomorrow's pollution cannot affect today's mortality, so any association between future pollution and current mortality indicates confounding by weather, seasonality, or other time-varying factors.

If the estimated effect of future pollution is similar in magnitude to the estimated effect of current pollution, this casts doubt on the causal interpretation of the primary estimate.

### 9.3.6 Empirical Example: Job Training and Labor Market Outcomes

In labor economics, negative control outcomes are often used to diagnose selection bias in job training evaluations. Suppose we study the effect of a job training program on earnings. A negative control outcome might be the participant's pre-treatment credit score or pre-treatment health status. The program cannot affect these pre-treatment variables, so any difference between treated and control groups reveals selection bias.

Similarly, a negative control exposure might be the participant's eligibility for a program that they did not actually attend. If eligibility for an unrelated program predicts earnings, this suggests that unmeasured characteristics drive both program participation and labor market outcomes.

### 9.3.7 Negative Controls as a Complement to Sensitivity Analysis

Negative controls do not replace formal sensitivity analysis. They do not quantify the size of hidden bias or provide bounds on the treatment effect. Instead, they serve as a qualitative diagnostic tool. They help answer questions such as:

- Is unmeasured confounding likely to be present?
- In what direction might it bias the estimate?
- Are treated and control groups comparable in ways not captured by observed covariates?

When negative controls show no association with treatment, this increases confidence in the study design. When they show strong associations, this signals that the primary estimate should be interpreted with caution.

### 9.3.8 Summary

Negative controls provide a simple but powerful diagnostic for unmeasured confounding. They rely on substantive knowledge to identify variables that should not be causally affected by the treatment. When such variables nevertheless show associations with treatment, this reveals the presence of hidden bias. Negative controls therefore complement the formal sensitivity analysis tools developed earlier, providing an empirical check on the plausibility of the no-unmeasured-confounding assumption.

## 9.4 Sensitivity analysis to unmeasured confounding

In randomized experiments, the assignment mechanism is known and controlled, so we can treat treatment as independent of all pre-treatment covariates, observed or unobserved. In observational studies, this luxury is gone. We adjust for observed covariates using matching, weighting, or regression, but we can never be sure that we have measured everything that matters. The central worry is *unmeasured confounding*: some unobserved variable may influence both treatment and outcome, biasing our estimates even after careful design.

Sensitivity analysis asks a very specific question: *how strong would an unmeasured confounder have to be, in order to explain away the observed treatment effect?* Rather than pretending that there is no unmeasured bias, we quantify how robust our conclusions are to departures from the idealized assumption of no hidden bias.

In this section, we develop a formal framework for sensitivity analysis following Rosenbaum's approach for matched observational studies. We begin with the simplest setting of matched pairs and a binary outcome or signed test statistic, then generalize to other outcomes and statistics. Throughout, the key idea is to replace the assumption of random assignment within matched sets by a *bounded departure* from randomization, indexed by a sensitivity parameter $\Gamma \geq 1$. We then study how $p$-values and confidence intervals change as $\Gamma$ increases.

## 9.5 Matched pairs and hidden bias

Consider an observational study with $N$ units, each with a binary treatment $Z_i \in \{0, 1\}$, an outcome $Y_i$, and observed covariates $X_i$. Suppose we have constructed a matched-pair design: the $N$ units are grouped into $I$ pairs, indexed by $i = 1, \ldots, I$, with two units per pair, indexed by $j = 1, 2$. Within each pair, the two units have similar observed covariates $X_{i1} \approx X_{i2}$, but may differ in treatment status $Z_{ij}$.

In a randomized experiment with matched pairs, we would have

$$\Pr(Z_{i1} = 1, Z_{i2} = 0) = \Pr(Z_{i1} = 0, Z_{i2} = 1) = \frac{1}{2},$$

and the treatment assignment within each pair would be independent of all covariates and potential outcomes. In an observational study, we instead imagine that treatment assignment within each pair may depend on an unobserved covariate $U_{ij}$ that we did not match on.

Let $Y_{ij}(1)$ and $Y_{ij}(0)$ denote the potential outcomes for unit $j$ in pair $i$ under treatment and control, respectively. We observe

$$Y_{ij}^{\text{obs}} = Z_{ij}Y_{ij}(1) + (1 - Z_{ij})Y_{ij}(0).$$

We are interested in testing a null hypothesis about treatment effects, for example the sharp null of no effect,

$$H_0 : Y_{ij}(1) = Y_{ij}(0) \quad \text{for all } i, j.$$

Under this sharp null, the observed outcomes do not depend on treatment, so we can impute all missing potential outcomes and treat the outcomes as fixed numbers. The only randomness comes from the treatment assignment within pairs.

In a randomized experiment, the probability that unit $j$ in pair $i$ receives treatment would be exactly 1/2. In an observational study with hidden bias, we instead posit that the odds of treatment may differ between the two units in a pair, but that this difference is bounded. This leads to Rosenbaum's $\Gamma$-model for hidden bias.

## 9.6 Rosenbaum's $\Gamma$-model for treatment assignment

Within pair $i$, let $\pi_{ij} = \Pr(Z_{ij} = 1 \mid X_{i1}, X_{i2}, U_{i1}, U_{i2})$ denote the conditional probability that unit $j$ receives treatment, given both observed and unobserved covariates. Because there is exactly one treated unit per pair, we have

$$\pi_{i1} + \pi_{i2} = 1.$$

In a randomized experiment, we would have $\pi_{i1} = \pi_{i2} = 1/2$. In an observational study, we allow $\pi_{i1}$ and $\pi_{i2}$ to differ, but we bound how different they can be. Specifically, we assume that there exists a parameter $\Gamma \geq 1$ such that, for each pair $i$,

$$\frac{1}{\Gamma} \leq \frac{\pi_{i1}/(1 - \pi_{i1})}{\pi_{i2}/(1 - \pi_{i2})} \leq \Gamma.$$

When $\Gamma = 1$, we recover random assignment within pairs: $\pi_{i1} = \pi_{i2} = 1/2$. As $\Gamma$ increases above 1, the treatment probabilities can become more imbalanced within pairs, reflecting stronger potential hidden bias.

This model can be motivated by a logistic treatment assignment model. Suppose that, within pair $i$,

$$\Pr(Z_{ij} = 1 \mid X_{ij}, U_{ij}) \propto \exp(\alpha X_{ij} + \gamma U_{ij}),$$

where $U_{ij}$ is an unobserved scalar covariate and $\gamma$ measures its influence on treatment. If the two units in a pair have the same observed covariates $X_{i1} = X_{i2}$, but may differ in $U_{ij}$, then the odds ratio of treatment between the two units is

$$\frac{\pi_{i1}/(1 - \pi_{i1})}{\pi_{i2}/(1 - \pi_{i2})} = \exp\{\gamma(U_{i1} - U_{i2})\}.$$

If we assume that $U_{ij}$ is bounded in such a way that its value is between 0 and 1, $|\gamma(U_{i1} - U_{i2})| \leq \log \Gamma$, then the odds ratio is bounded between $1/\Gamma$ and $\Gamma$, which is equivalent to the constraint above. Thus, $\Gamma$ can be interpreted as the maximum factor by which an unmeasured covariate can multiply the odds of treatment within a matched pair.

The key point is that $\Gamma$ is not estimated from the data; it is a sensitivity parameter that we vary. For each fixed $\Gamma$, we ask: *among all treatment assignment mechanisms consistent with this bound on hidden bias, what is the worst-case p-value for our test statistic?* If our conclusions remain unchanged up to a reasonably large $\Gamma$, we say that the study is insensitive to moderate hidden bias. If a small $\Gamma$ suffices to overturn our conclusions, the study is highly sensitive.

## 9.7 Signed rank statistics in matched pairs

To make these ideas concrete, consider a matched-pair study with a continuous outcome and the sharp null of no effect. For each pair $i$, define the treated-minus-control difference in observed outcomes,

$$D_i = (Z_{i1} - Z_{i2})\, Y_{i1}^{\text{obs}} + (Z_{i2} - Z_{i1})\, Y_{i2}^{\text{obs}}.$$

Under the sharp null, this is simply the difference in outcomes between the two units in the pair, with a sign determined by which unit received treatment. Many test statistics can be built from the $D_i$'s. A simple choice is the *sign test*, which uses only the sign of each difference:

$$T = \sum_{i=1}^{I} S_i, \qquad S_i = \mathbf{1}\{D_i > 0\}.$$

More powerful tests, such as the Wilcoxon signed-rank test, weight the signs by the ranks of the absolute differences. For sensitivity analysis, the key property is that the test statistic can be written as a sum of contributions from each pair,

$$T = \sum_{i=1}^{I} w_i S_i,$$

where $w_i \geq 0$ are fixed, known weights (for the sign test, $w_i = 1$; for the signed-rank test, $w_i$ is the rank of $|D_i|$). Under the sharp null, the $w_i$ are fixed functions of the observed data, and the randomness in $T$ comes solely from the random signs $S_i$, which depend on which unit in each pair is treated.

In a randomized experiment, each pair has $\Pr(S_i = 1) = \Pr(S_i = 0) = 1/2$, so $T$ is a sum of independent Bernoulli or weighted Bernoulli variables with known distribution. In an observational study with hidden bias, the probability that the treated unit has the larger outcome may differ between pairs and may deviate from 1/2. The $\Gamma$-model bounds these probabilities.

## 9.8 Worst-case $p$-values under hidden bias

Under the sharp null and the $\Gamma$-model, the probability that the treated unit in pair $i$ has the larger outcome is bounded between

$$\frac{1}{1+\Gamma} \leq \Pr(S_i = 1 \mid \text{data}) \leq \frac{\Gamma}{1+\Gamma}.$$

The exact value may differ from pair to pair and is unknown, because it depends on the unobserved $U_{ij}$. However, we can derive *worst-case* bounds on the distribution of $T$ by choosing, for each pair, the value of $\Pr(S_i = 1)$ that makes the observed test statistic as likely as possible under the null.

For a one-sided test in which large values of $T$ are evidence against the null, the worst-case scenario (most favorable to the null) is to make $S_i = 1$ as unlikely as possible in pairs where $w_i$ is large. Intuitively, if a pair has a large weight $w_i$, then a positive sign contributes a lot to $T$, so the null hypothesis would like that sign to be as improbable as allowed by $\Gamma$. Formally, one can show that the worst-case distribution of $T$ under the null and the $\Gamma$-model is obtained by setting

$$\Pr(S_i = 1 \mid \text{data}) = \begin{cases} \frac{1}{1+\Gamma}, & \text{if } w_i > 0, \\ \text{any value in } \left[\frac{1}{1+\Gamma}, \frac{\Gamma}{1+\Gamma}\right], & \text{if } w_i = 0. \end{cases}$$

In practice, for signed-rank or sign tests, all $w_i > 0$, so we take $\Pr(S_i = 1) = 1/(1+\Gamma)$ for all pairs. Then, under the null and the worst-case hidden bias consistent with $\Gamma$, the test statistic $T$ is stochastically dominated by a sum of independent Bernoulli variables with success probability $1/(1+\Gamma)$. This yields an upper bound on the one-sided $p$-value:

$$p_\Gamma^+ = \Pr_\Gamma(T^* \geq T_{\text{obs}}),$$

where $T^*$ is a random variable with the worst-case distribution under the null and $\Gamma$, and $T_{\text{obs}}$ is the observed value of the test statistic.

Similarly, for a one-sided test in which small values of $T$ are evidence against the null, the worst-case scenario is to make $S_i = 1$ as likely as possible, i.e., $\Pr(S_i = 1) = \Gamma/(1+\Gamma)$. For two-sided tests, one can combine the two one-sided bounds or use more refined arguments.

In large samples, the distribution of $T$ under the worst-case model can be approximated by a normal distribution using the central limit theorem. For example, for the sign test with $w_i = 1$, we have

$$E_\Gamma[T] = I \cdot \frac{\Gamma}{1+\Gamma}, \qquad \text{Var}_\Gamma(T) = I \cdot \frac{1}{1+\Gamma}\left(1 - \frac{\Gamma}{1+\Gamma}\right),$$

under the worst-case model for large $T$. Then

$$Z_\Gamma = \frac{T_{\text{obs}} - E_\Gamma[T]}{\sqrt{\text{Var}_\Gamma(T)}}$$

is approximately standard normal under the null and the worst-case hidden bias, and the one-sided $p$-value is approximately

$$p_\Gamma^+ \approx 1 - \Phi(Z_\Gamma),$$

where $\Phi$ is the standard normal distribution function.

By computing $p_\Gamma$ for increasing values of $\Gamma$, we obtain a *sensitivity curve*: a function $\Gamma \mapsto p_\Gamma$ that shows how the evidence against the null weakens as we allow for stronger hidden bias. The *design sensitivity* of a study is the largest value of $\Gamma$ for which the conclusion (e.g., rejection of the null at level 0.05) still holds.

## 9.9 Interpreting $\Gamma$ in practice

The parameter $\Gamma$ has a concrete interpretation in terms of the odds of treatment within matched pairs. Recall that, under the $\Gamma$-model,

$$\frac{1}{\Gamma} \leq \frac{\pi_{i1}/(1-\pi_{i1})}{\pi_{i2}/(1-\pi_{i2})} \leq \Gamma,$$

where $\pi_{ij}$ is the probability that unit $j$ in pair $i$ receives treatment, given both observed and unobserved covariates. Thus, $\Gamma$ is the maximum factor by which an unmeasured covariate can multiply the odds of treatment between two otherwise similar units.

For example, $\Gamma = 1.2$ means that, within a matched pair, one unit could be up to 20% more likely to receive treatment than the other because of an unmeasured covariate. $\Gamma = 2$ means that one unit could be twice as likely to receive treatment as the other. If a study's conclusions are insensitive up to $\Gamma = 2$, then even if there were an unmeasured covariate that doubled the odds of treatment within matched pairs, the observed effect would still be statistically significant at the chosen level.

Interpreting $\Gamma$ requires substantive knowledge. In some settings, a factor of 1.2 in the odds of treatment may be implausibly small; in others, a factor of 2 may be quite modest. The value of sensitivity analysis is not that it tells us whether there is hidden bias, but that it makes explicit the strength of hidden bias required to overturn our conclusions.

## 9.10 Sensitivity analysis for confidence intervals

So far, we have focused on sensitivity analysis for hypothesis tests. We can also perform sensitivity analysis for confidence intervals, particularly for the Hodges–Lehmann point estimate of a treatment effect in matched pairs.

Suppose we are interested in a constant additive treatment effect $\tau$, so that

$$Y_{ij}(1) = Y_{ij}(0) + \tau \quad \text{for all } i, j.$$

Under this model, the sharp null $H_0 : \tau = \tau_0$ implies that

$$Y_{ij}^{\text{obs}} - \tau_0 Z_{ij} = Y_{ij}(0),$$

so we can subtract $\tau_0$ from the treated outcomes and treat the adjusted outcomes as if the null of no effect held. For each candidate value $\tau_0$, we can perform a sensitivity analysis test of the null $H_0 : \tau = \tau_0$ under the $\Gamma$-model, obtaining a $p$-value $p_\Gamma(\tau_0)$. The $(1-\alpha)$-level confidence set for $\tau$ under sensitivity parameter $\Gamma$ is then

$$C_\Gamma(1-\alpha) = \{\tau_0 : p_\Gamma(\tau_0) > \alpha\}.$$

In practice, one often uses the signed-rank statistic applied to the adjusted pair differences

$$D_i(\tau_0) = D_i - \tau_0,$$

and computes the worst-case $p$-value under the $\Gamma$-model for each $\tau_0$. The resulting confidence interval widens as $\Gamma$ increases, reflecting the additional uncertainty due to possible hidden bias. At $\Gamma = 1$, we recover the usual randomization-based confidence interval. As $\Gamma$ grows, the interval may eventually include zero, indicating that, under sufficiently strong hidden bias, the data are compatible with no effect.

This procedure yields a *sensitivity interval*: a family of confidence intervals indexed by $\Gamma$. Reporting such intervals allows readers to see how the estimated effect and its uncertainty change as we allow for stronger unmeasured confounding.

## 9.11 Beyond matched pairs: matched sets and other statistics

The basic ideas extend beyond matched pairs to matched sets of size $m_i \geq 2$, where each set contains one treated unit and $m_i - 1$ controls. In that case, the $\Gamma$-model bounds the treatment probabilities within each set:

$$\frac{1}{\Gamma} \leq \frac{\pi_{ij}/(1-\pi_{ij})}{\pi_{ik}/(1-\pi_{ik})} \leq \Gamma \quad \text{for all } j, k \text{ in set } i,$$

where $\pi_{ij}$ is the probability that unit $j$ in set $i$ is the treated unit. Under the sharp null, we can again construct test statistics as sums of contributions from each set, and derive worst-case $p$-values under the $\Gamma$-model. The algebra is more involved, but the conceptual structure is the same: we bound the departure from random assignment within matched sets and compute how this bound affects the distribution of our test statistic.

Similarly, sensitivity analysis can be applied to other test statistics, such as differences in means, rank-based statistics, or statistics based on regression residuals, provided that we can characterize their distribution under the null and the $\Gamma$-model. Rosenbaum's framework is quite general and can be adapted to many designs and estimands.

## 9.12 Amplification and multiple unmeasured covariates

One might worry that the $\Gamma$-model is too simple: it effectively assumes a single unmeasured covariate that affects treatment assignment. In reality, there may be many unmeasured factors. Rosenbaum introduces the idea of *amplification*: re-expressing a given $\Gamma$ in terms of a joint effect of multiple unmeasured covariates on treatment and outcome.

The basic idea is that a given $\Gamma$ can be achieved by many combinations of unmeasured covariates, each with smaller effects. For example, suppose there are two unmeasured covariates, $U^{(1)}$ and $U^{(2)}$, each of which multiplies the odds of treatment by at most $\Gamma_1$ and $\Gamma_2$, respectively. If their effects on treatment are multiplicative and independent, the combined effect on the odds of treatment is at most $\Gamma_1\Gamma_2$. Thus, a sensitivity parameter $\Gamma = 2$ could correspond to a single unmeasured covariate that doubles the odds of treatment, or to two unmeasured covariates that each increase the odds by a factor of $\sqrt{2}$, and so on.

Amplification allows us to interpret a given $\Gamma$ in terms of more nuanced stories about unmeasured confounding. It also emphasizes that sensitivity analysis is not about a single hidden variable, but about the overall strength of unmeasured bias.

## 9.13 Design sensitivity and power under hidden bias

In large samples, the power of a sensitivity analysis depends on both the true treatment effect and the design of the study. Rosenbaum introduces the concept of *design sensitivity*: the largest value of $\Gamma$ for which a given test will reject the null with probability tending to one as the sample size grows, under a fixed alternative.

Intuitively, some designs are more robust to hidden bias than others. For example, matching on strong predictors of treatment and outcome, using fine balance on key covariates, and avoiding extrapolation can all increase design sensitivity. Conversely, poor overlap, weak covariates, or highly variable outcomes can reduce design sensitivity. Sensitivity analysis thus connects naturally to design: we can compare alternative designs not only in terms of covariate balance and variance, but also in terms of how robust they are to unmeasured confounding.

### 9.14 A narrative example

To make the ideas more concrete, imagine a study of the effect of a job training program on earnings. We match treated and control individuals on age, education, prior earnings, and region, forming 500 matched pairs. A signed-rank test of the null of no effect yields a $p$-value of $p_1 = 0.001$ under $\Gamma = 1$, suggesting a strong positive effect of the program.

We then perform a sensitivity analysis. At $\Gamma = 1.2$, the worst-case one-sided $p$-value is $p_{1.2} = 0.005$; at $\Gamma = 1.5$, it is $p_{1.5} = 0.03$; at $\Gamma = 2.0$, it is $p_{2.0} = 0.12$. Thus, the conclusion of a statistically significant positive effect at the 5% level is insensitive to hidden bias up to $\Gamma \approx 1.5$, but could be overturned by an unmeasured covariate that doubles the odds of treatment within matched pairs.

Substantively, we ask: is it plausible that, among individuals matched on age, education, prior earnings, and region, there exists an unmeasured factor (such as motivation or social support) that both doubles the odds of enrolling in the program and is strongly related to earnings? If yes, then we should be cautious in interpreting the estimated effect as causal. If no, then the sensitivity analysis strengthens our confidence that the observed effect reflects a real causal impact of the program.

### 9.15 What sensitivity analysis does and does not do

Sensitivity analysis does not prove that there is no unmeasured confounding. It does not produce a single number that tells us whether a study is "valid". Instead, it provides a structured way to ask: *how large would unmeasured bias have to be to materially change our conclusions?* This is a more modest, but also more honest, goal.

The value of sensitivity analysis lies in its transparency. It forces us to articulate the strength of hidden bias required to explain away our findings, and to confront whether such bias is plausible in the substantive context. It also connects design and analysis: better designs yield higher design sensitivity, making conclusions more robust to unmeasured confounding.

In practice, sensitivity analysis should be reported alongside covariate balance diagnostics and primary estimates. Together, these pieces form a coherent narrative: we show that the design balances observed covariates, that the estimated effect is substantial and precisely estimated, and that the conclusions are insensitive to reasonably large departures from the idealized assumption of no hidden bias.

## Appendix: Some Mathematical and Computational Foundations of Sensitivity Analysis

This appendix develops the mathematical structure that underlies the R demonstrations for Rosenbaum's sensitivity analysis. The goal is to present a complete and self-contained treatment of the test statistics, probability models, and worst-case bounds that appear when we analyze matched observational studies under the possibility of unmeasured confounding. The exposition follows the spirit of Rosenbaum's *Observational Studies* (2002), but is adapted to the matched-pair designs constructed earlier using `approxmatch`. The intention is that a student who reads this appendix should be able to understand exactly what the R code is doing, why the formulas take the shape they do, and how the sensitivity parameter $\Gamma$ enters the analysis.

## A.1 Matched Pairs and the Sharp Null

We begin with a matched-pair design consisting of $I$ pairs, each containing one treated unit and one control unit. For pair $i$, let the two units be indexed by $j = 1, 2$. Each unit has two potential outcomes, $Y_{ij}(1)$ under treatment and $Y_{ij}(0)$ under control, and we observe

$$Y_{ij}^{\text{obs}} = Z_{ij}Y_{ij}(1) + (1 - Z_{ij})Y_{ij}(0),$$

where $Z_{ij} \in \{0, 1\}$ indicates treatment. The sharp null hypothesis of no treatment effect,

$$H_0 : Y_{ij}(1) = Y_{ij}(0) \quad \text{for all } i, j,$$

implies that the observed outcomes do not depend on treatment. Under this null, the missing potential outcomes are known, and the only randomness in the problem comes from the treatment assignment mechanism.

For each pair, we define the treated-minus-control difference

$$D_i = (Z_{i1} - Z_{i2})Y_{i1}^{\text{obs}} + (Z_{i2} - Z_{i1})Y_{i2}^{\text{obs}}.$$

Under the sharp null, this is simply the difference in outcomes between the two units in the pair, with a sign determined by which unit received treatment. The collection $\{D_i\}_{i=1}^{I}$ therefore contains all the information needed for randomization-based inference.

## A.3 The Sign Test Under Hidden Bias

The simplest statistic for matched pairs is the sign statistic. For each pair, define

$$S_i = \mathbf{1}\{D_i > 0\},$$

which records whether the treated unit had a larger outcome than its control partner. The sign test statistic is

$$T = \sum_{i=1}^{I} S_i.$$

Under random assignment ($\Gamma = 1$), each $S_i$ is a Bernoulli random variable with probability $1/2$, so $T$ follows a Binomial$(I, 1/2)$ distribution. Under hidden bias, however, the probability of a positive sign may differ from $1/2$. The worst-case scenario for the null hypothesis is the one that makes the observed statistic as likely as possible. Under the $\Gamma$-model, this occurs when

$$\Pr(S_i = 1) = \frac{\Gamma}{1 + \Gamma},$$

the smallest value allowed by the model. In this case,

$$T \sim \text{Binomial}\left(I, \frac{\Gamma}{1 + \Gamma}\right).$$

The one-sided $p$-value for testing whether the treated units tend to have larger outcomes is

$$p_\Gamma^+ = \Pr(T^* \geq T_{\text{obs}}), \qquad T^* \sim \text{Binomial}\left(I, \frac{\Gamma}{1 + \Gamma}\right).$$

As $\Gamma$ increases, the distribution of $T^*$ shifts to the right, and the $p$-value increases. The sensitivity value $\Gamma^*$ is the smallest value of $\Gamma$ for which $p_\Gamma^+$ exceeds the chosen significance level (typically 0.05). This value quantifies how strong the unmeasured confounding would need to be to explain away the observed directional effect.

## A.4 The Wilcoxon Signed-Rank Test Under Hidden Bias

The sign test uses only the direction of each pair difference and ignores its magnitude. The Wilcoxon signed-rank test incorporates both direction and magnitude by ranking the absolute differences. Let $R_i$ be the rank of $|D_i|$ among the $I$ pairs. The Wilcoxon statistic is

$$W = \sum_{i=1}^{I} R_i S_i.$$

Under random assignment ($\Gamma = 1$), each $S_i$ is Bernoulli(1/2), and the expectation and variance of $W$ are

$$E[W] = \frac{1}{2}\sum_{i=1}^{I} R_i, \qquad \mathrm{Var}(W) = \frac{1}{4}\sum_{i=1}^{I} R_i^2.$$

Under hidden bias with sensitivity parameter $\Gamma$, the worst-case probability of a positive sign is again $1/(1+\Gamma)$, so

$$E_\Gamma[W] = \frac{\Gamma}{1+\Gamma}\sum_{i=1}^{I} R_i, \qquad \mathrm{Var}_\Gamma(W) = \frac{\Gamma}{(1+\Gamma)^2}\sum_{i=1}^{I} R_i^2.$$

A normal approximation yields the worst-case $p$-value:

$$Z_\Gamma = \frac{W_{\mathrm{obs}} - E_\Gamma[W]}{\sqrt{\mathrm{Var}_\Gamma(W)}}, \qquad p_\Gamma^+ = 1 - \Phi(Z_\Gamma).$$

As with the sign test, the sensitivity value $\Gamma^*$ is the point at which $p_\Gamma^+$ crosses the significance threshold.

## A.5 Hodges–Lehmann Sensitivity Intervals

The sign and Wilcoxon tests provide sensitivity analysis for hypothesis testing. To obtain sensitivity intervals for the treatment effect, we adopt the constant additive effect model

$$Y_{ij}(1) = Y_{ij}(0) + \tau.$$

Under this model, the adjusted pair differences are

$$D_i(\tau_0) = D_i - \tau_0.$$

For each candidate value $\tau_0$, we test the sharp null $H_0 : \tau = \tau_0$ using the sign test under the $\Gamma$-model. The $(1-\alpha)$ sensitivity interval is the set of all $\tau_0$ for which the worst-case $p$-value exceeds $\alpha$. As $\Gamma$ increases, the interval widens, reflecting the increased uncertainty due to possible hidden bias.

## A.6 Sensitivity Curves and the Interpretation of $\Gamma^*$

A convenient way to summarize the results of a sensitivity analysis is to plot the worst-case $p$-value as a function of $\Gamma$. This *sensitivity curve* begins at the usual randomization $p$-value when $\Gamma = 1$ and increases monotonically as $\Gamma$ grows. The sensitivity value $\Gamma^*$ is the point at which the curve crosses the significance threshold.

The interpretation of $\Gamma^*$ is straightforward. If $\Gamma^* = 1.5$, for example, then an unmeasured covariate that increased the odds of treatment within a matched pair by a factor of 1.5 could explain away the observed effect. If such a covariate seems implausible in the substantive context, the study may be considered robust. If it seems entirely plausible, the study is sensitive to unmeasured confounding.

## A.7 Concluding Remarks

The sensitivity analysis developed here provides a principled way to quantify the impact of unmeasured confounding in matched observational studies. Rather than assuming that matching has eliminated all bias, we explicitly model the possibility of hidden bias and determine how strong it would need to be to overturn our conclusions. The resulting sensitivity values and intervals offer a transparent and interpretable measure of robustness, complementing the design and analysis tools developed earlier in the course.